islands and thin filaments of *background pixels*. These techniques are useful for handling noisy images where some pixels have the wrong binary value. For instance, it might be known that an object cannot contain a "hole", or that the object's border must be smooth.

Figure 25-11 shows an example of morphological processing. Figure (a) is the binary image of a fingerprint. Algorithms have been developed to analyze these patterns, allowing individual fingerprints to be matched with those in a database. A common step in these algorithms is shown in (b), an operation called **skeletonization**. This simplifies the image by *removing* redundant pixels; that is, changing appropriate pixels from black to white. This results in each ridge being turned into a line only a single pixel wide.

Tables 25-1 and 25-2 show the skeletonization program. Even though the fingerprint image is binary, it is held in an array where each pixel can run from 0 to 255. A black pixel is denoted by 0, while a white pixel is denoted by 255. As shown in Table 25-1, the algorithm is composed of 6 iterations that gradually erode the ridges into a thin line. The number of iterations is chosen by trial and error. An alternative would be to stop when an iteration makes no changes.

During an iteration, each pixel in the image is evaluated for being *removable*; the pixel meets a set of criteria for being changed from black to white. Lines 200-240 loop through each pixel in the image, while the subroutine in Table 25-2 makes the evaluation. If the pixel under consideration is not removable, the subroutine does nothing. If the pixel is removable, the subroutine changes its value from 0 to 1. This indicates that the pixel is still black, but will be changed to white at the end of the iteration. After all the pixels have been evaluated, lines 260-300 change the value of the marked pixels from 1 to 255. This two-stage process results in the thick ridges being eroded equally from all directions, rather than a pattern based on how the rows and columns are scanned.

a. Original fingerprint

b. Skeletonized fingerprint

FIGURE 25-11
Binary skeletonization. The binary image of a fingerprint, (a), contains ridges that are many pixels wide. The skeletonized version, (b), contains ridges only a single pixel wide.